

# ZACHARY A. BOLTON

Full-Stack & Machine Learning Engineer



Phone: +1-770-334-9062

Email: [zacbolton2129@gmail.com](mailto:zacbolton2129@gmail.com)



ZacharyABolton



ZacABolton



Portfolio

I build products end to end — from years leading technology at an EdTech company to engineering Boostlingo's real-time voice AI in production. I pair product pragmatism with rigorous ML methodology: eval harnesses, competitive benchmarks, and a publicly released accuracy study. Happiest taking ambiguous zero-to-one problems from architecture to deploy.

## EXPERIENCE

### BOOSTLINGO - August 2025 to Present

#### SENIOR MACHINE LEARNING ENGINEER

Language-services technology company delivering human and AI interpreting to healthcare, legal, and business clients worldwide.

- Lead engineer on Boostlingo's real-time AI Interpreter (.NET/Orleans): built voice-activity detection, crosstalk handling, and conversation-management mechanics for live interpreted calls serving ~35,000 minutes/month
- Built the team's LLM evaluation infrastructure from scratch: translation-quality harness with GEMBA scoring, multilingual datasets and baselines, wired into CI — enabling confident, monitored expansion to 9 new language pairs
- Designed, conducted, and authored **publicly released AI interpreting accuracy study** benchmarking medically complex and legal scenarios against competing systems
- Drove hallucination-reduction and correct-language-output initiatives, cutting hallucination/character-break rates in the hardest edge cases from 56% to 5%; owned production deploys, feature-flagged rollouts, IVR/telephony integrations
- Evaluated PII-redaction vendors (Private.AI, CaseGuard, AssemblyAI) for sensitive healthcare call data

Stack: C#/.NET (Microsoft Orleans), Python, LLM & speech APIs (Gemini et al.), Twilio, Datadog LLM Observability, Git/CI-CD

### UNIVERSITY OF LONDON - April 2024 to April 2025

#### INDEPENDENT ML RESEARCHER

- Implemented **Siamese Sentence-BERT authorship-verification architecture** (AUC 0.900, F1 0.785 on VALLA benchmark vs PAN 2020 participants); developed novel bias-controlled LILA dataset achieving AUC/F1/C@1 > 0.99
- Deployed end-to-end data-processing/ML pipeline to AWS EC2 handling 10M+ words

Stack: PyTorch, HuggingFace Transformers, Python Scientific Stack, AWS EC2/S3

### KID CHESS, INC. - January 2015 to August 2025

#### SOFTWARE ENGINEER AND DIRECTOR OF TECHNOLOGY

- Built and led **full-stack online learning platform** from its 2020 launch (Angular, TypeScript, Azure), serving ~4,000 students; created GPT-4-powered personalized feedback system
- Automated business processes with Python pipelines (40% efficiency gain); led pandemic transition to fully online services, achieving 90% student retention

Stack: Angular, TypeScript, Azure, Python, OpenAI API, SQL

## EDUCATION

- **University of London:** First Class Honors BSc in Computer Science (Machine Learning & AI) | GPA: 90% | April 2025
- **Mainsail Partners:** **LEAP Leadership Acceleration Program** — Manager Training | June 2026

#### ML/AI Skills:

- Real-time Voice AI
- LLM Evals & Benchmarking
- PyTorch & HuggingFace
- NLP & Speech Processing
- LLM Observability
- Experiment Design

#### Programming:

- Python
- TypeScript/JavaScript
- C#/.NET (Orleans)
- Angular
- SQL
- Azure & AWS

#### Soft Skills:

- 0→1 Product Ownership
- Technical Writing
- Cross-team Collaboration
- Project Leadership
- Remote Collaboration